

AWS CloudTrail Lab: Monitor AWS Activity Using Event History and Trails

Summary:-

In this lab, **AWS CloudTrail** was used to monitor and audit activities performed within an AWS account. An **IAM** user with **Amazon EC2 Full Access** permissions was created to perform actions on an existing **Amazon EC2** instance. The recorded API events were reviewed using **CloudTrail Event History**, and details such as the user identity, event time, source IP address, and affected resources were verified. Finally, a CloudTrail **Trail** was configured to continuously deliver event logs to an **Amazon S3** bucket for long-term storage.

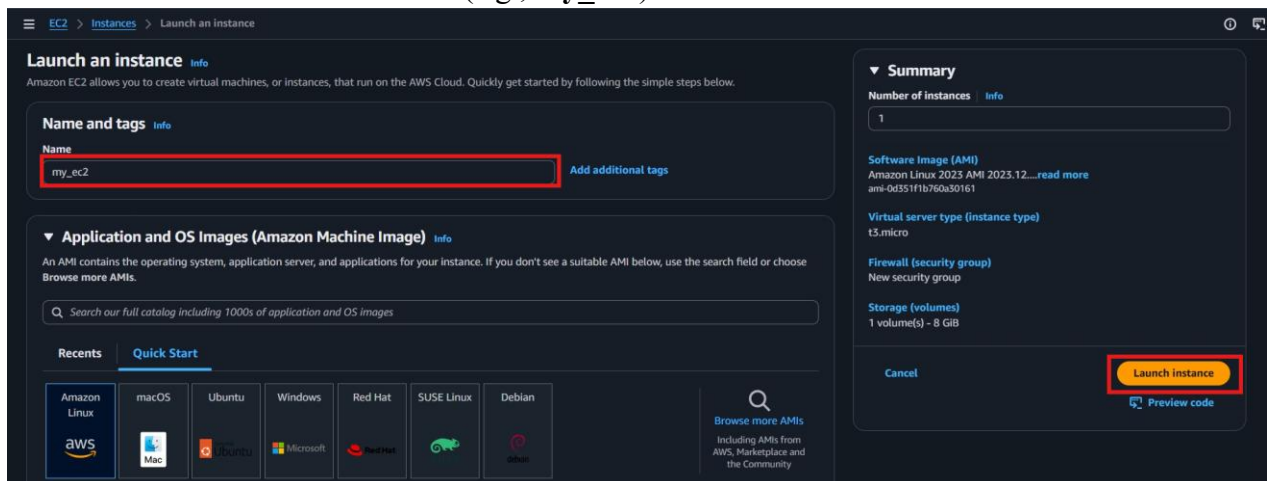
Let's Start:-

- **AWS CloudTrail** records API activity performed through the AWS Management Console, AWS CLI, SDKs, and other AWS services.
- **Event History** stores the last **90 days** of management events at no additional configuration.
- A **Trail** continuously delivers CloudTrail events to an **Amazon S3** bucket for long-term storage and auditing.

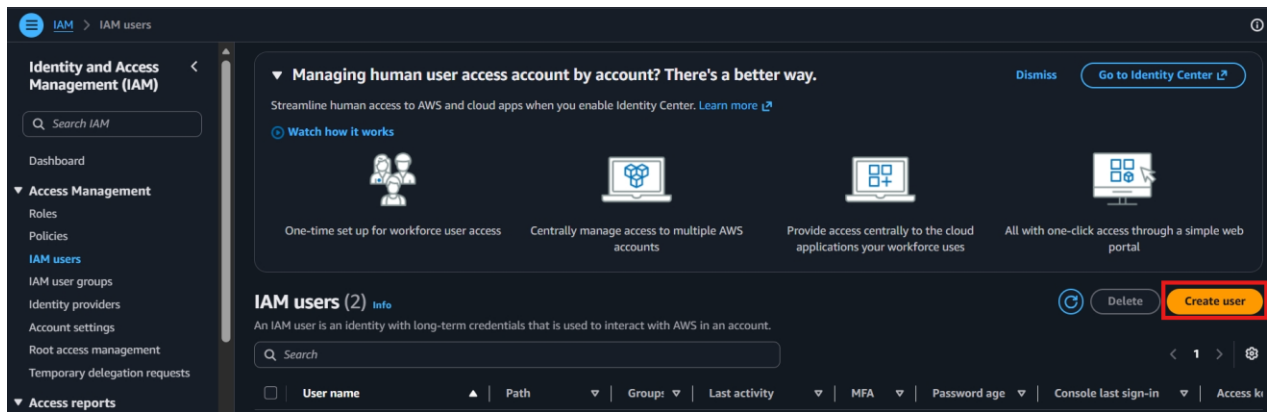
Lab Steps:-

Step 1 – Create an IAM User

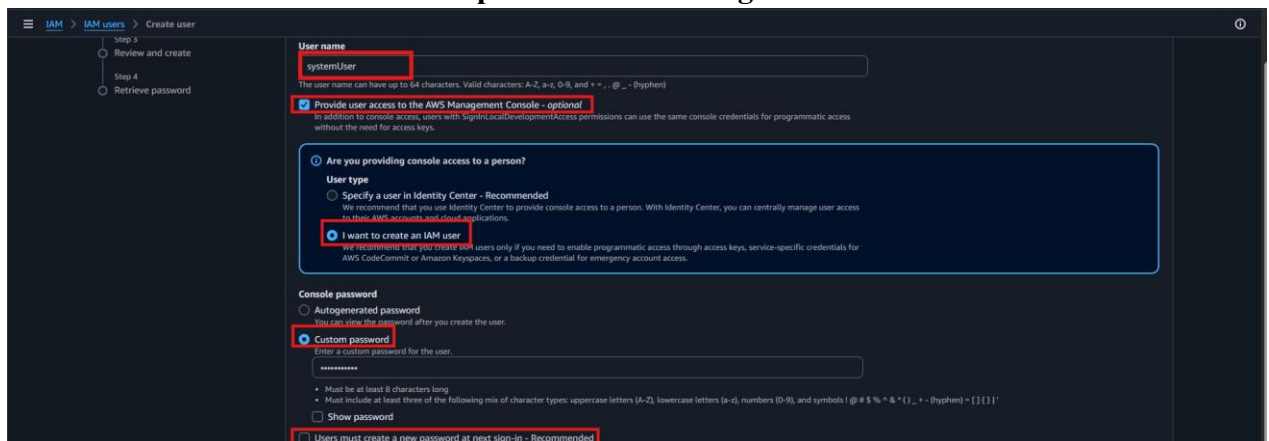
- Sign in to the **AWS Management Console** as the root user.
- Navigate to **Amazon EC2**.
- Launch a new EC2 instance (e.g., my_ec2).



- Navigate to **IAM**.
- Click **Users** → **Create User**.

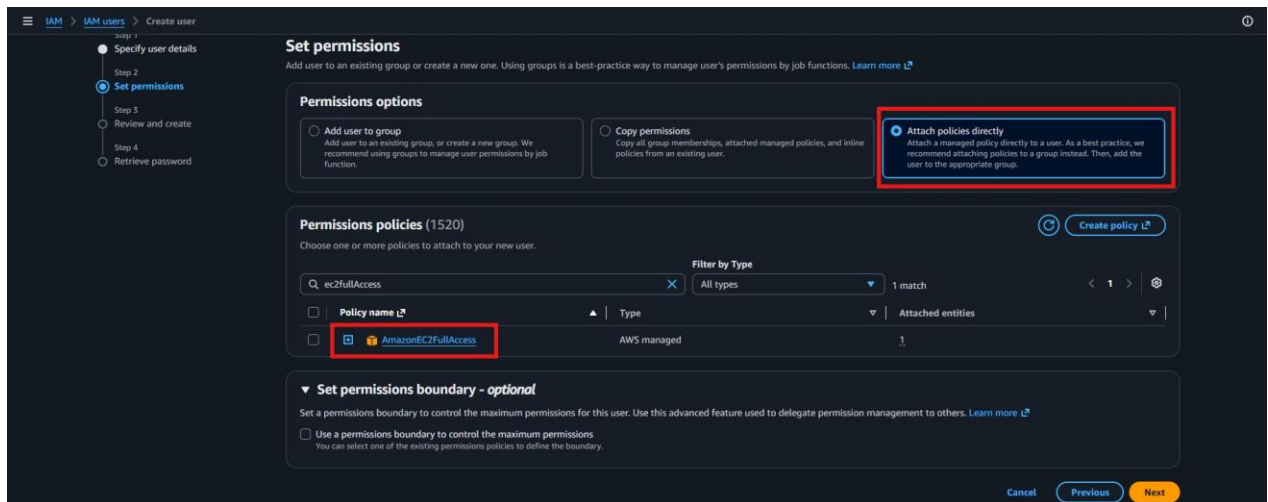


- Enter a user name.
- Enable **AWS Management Console Access**.
- Configure a custom password.
- Clear **User must create a new password at next sign-in**.



Step 2 – Assign EC2 Permissions

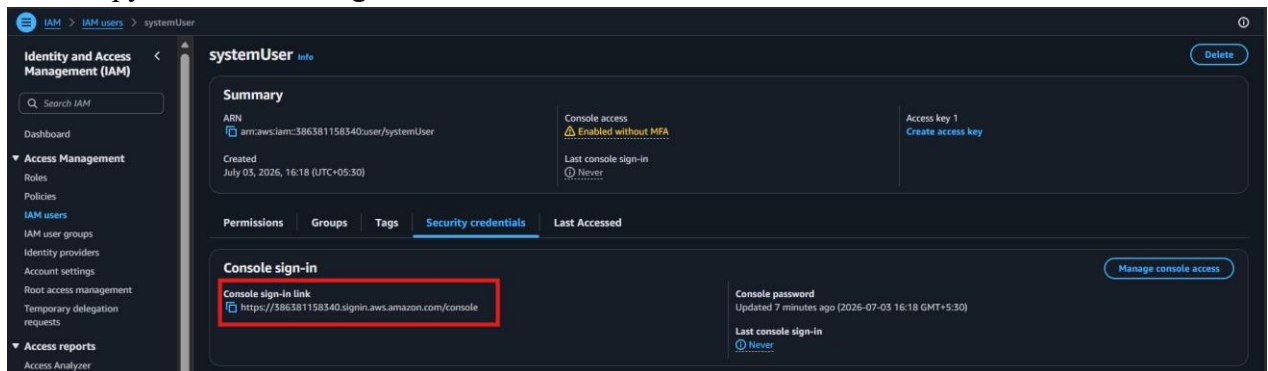
- Select **Attach Policies Directly**.
- Search for **AmazonEC2FullAccess**.
- Select the policy.



- Click **Create User**.

Step 3 – Sign In Using the IAM User

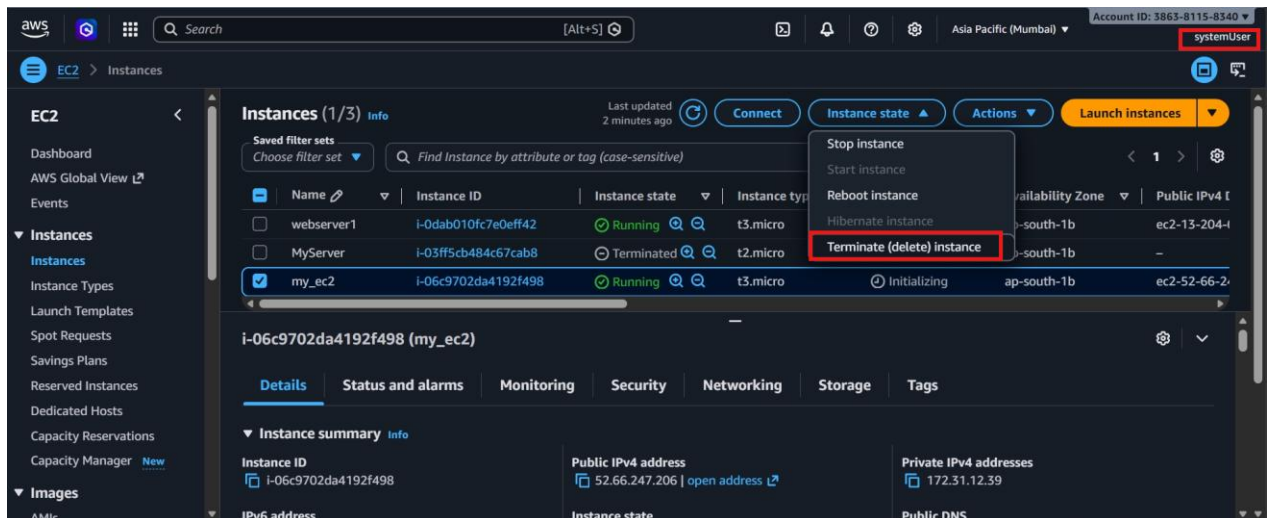
- Copy the IAM user sign-in URL.



- Open a new browser or an incognito window.
- Sign in using the newly created IAM user credentials.

Step 4 – Perform an EC2 Action

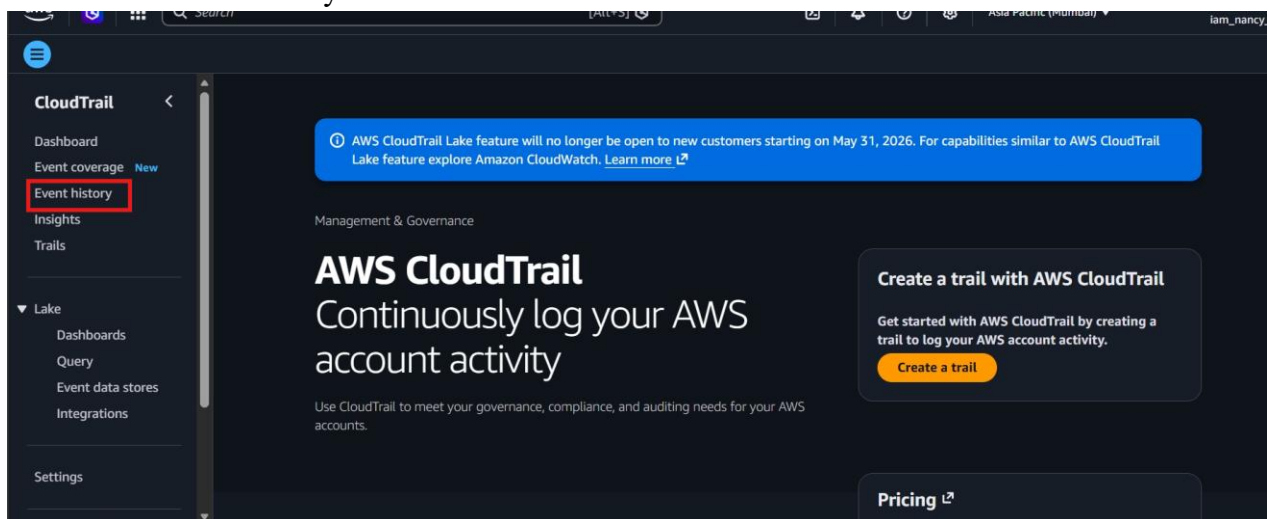
- Navigate to Amazon EC2.
- Select an existing EC2 instance.
- Click Instance State → Terminate Instance.



- Confirm the termination request.
(Attach Screenshot)

Step 5 – View CloudTrail Event History

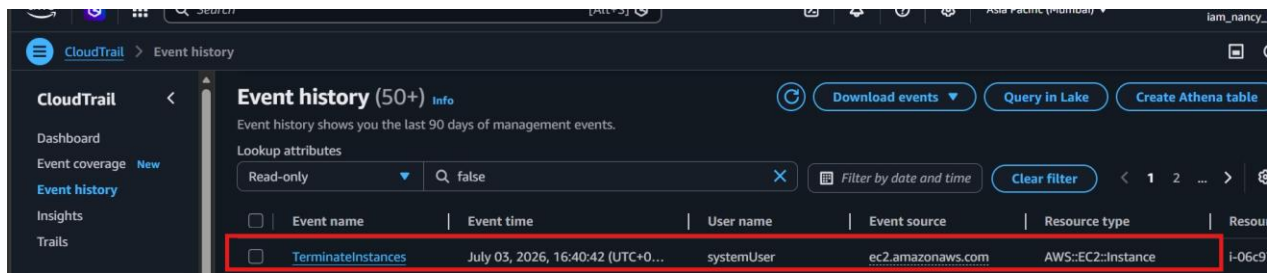
- Return to the root user session.
- Navigate to AWS CloudTrail.
- Select Event History.



- Locate the TerminateInstances event.
- Verify the recorded event details.

Step 6 – Review Event Details

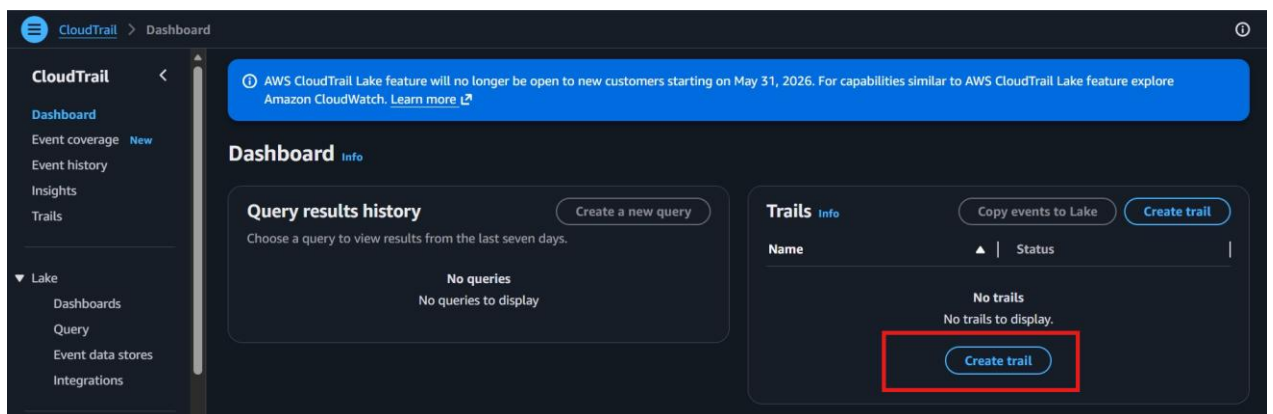
- Open the recorded event.



- Verify the following information:
 - IAM user who performed the action.
 - Event name.
 - Event time.
 - Source IP address.
 - EC2 Instance ID.
 - API request details.
- (Attach Screenshot)

Step 7 – Create a CloudTrail Trail

- Navigate to Trails.
- Click Create Trail.



- Enter a trail name (e.g., Trail1).
- Select Create New S3 Bucket.

CloudTrail > Dashboard > Create trail

Step 1: Choose trail attributes

Choose trail attributes

General details
A trail created in the console is a multi-region trail. [Learn more](#)

Trail name
Enter a display name for your trail.

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization
To review accounts in your organization, open AWS Organizations. [See all accounts](#)

Storage location [Info](#)

☒ Create new S3 bucket
Create a bucket to store logs for the trail.

☐ Use existing S3 bucket
Choose an existing bucket to store logs for this trail.

Trail log bucket and folder
Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.

Logs will be stored in aws-cloudtrail-logs-386381158340-c22420d9/AWSLogs/386381158340

- Keep the default log delivery settings.

Step 8 – Configure Encryption

- Review the default encryption settings.
- Verify that Amazon S3 Server-Side Encryption (SSE-S3) is enabled by default.
- (Optional) Select AWS KMS if stronger encryption is required.

CloudTrail > Dashboard > Create trail

☒ Create new S3 bucket
Create a bucket to store logs for the trail.

☐ Use existing S3 bucket
Choose an existing bucket to store logs for this trail.

Trail log bucket and folder
Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.

Logs will be stored in aws-cloudtrail-logs-386381158340-c22420d9/AWSLogs/386381158340

Log file SSE-KMS encryption [Info](#)

☒ Enabled

Additional settings

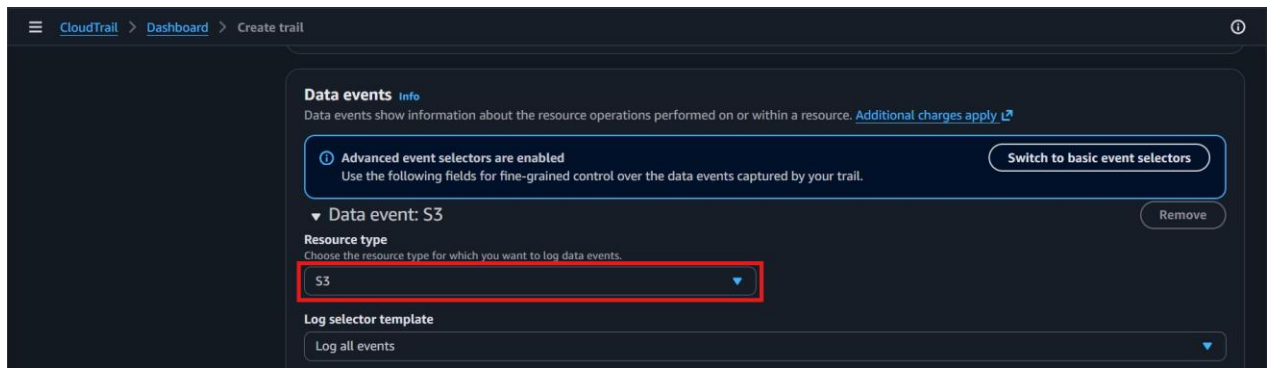
Log file validation [Info](#)
☒ Enabled

SNS notification delivery [Info](#)
☐ Enabled

CloudWatch Logs - optional

Step 9 – Enable Log File Validation

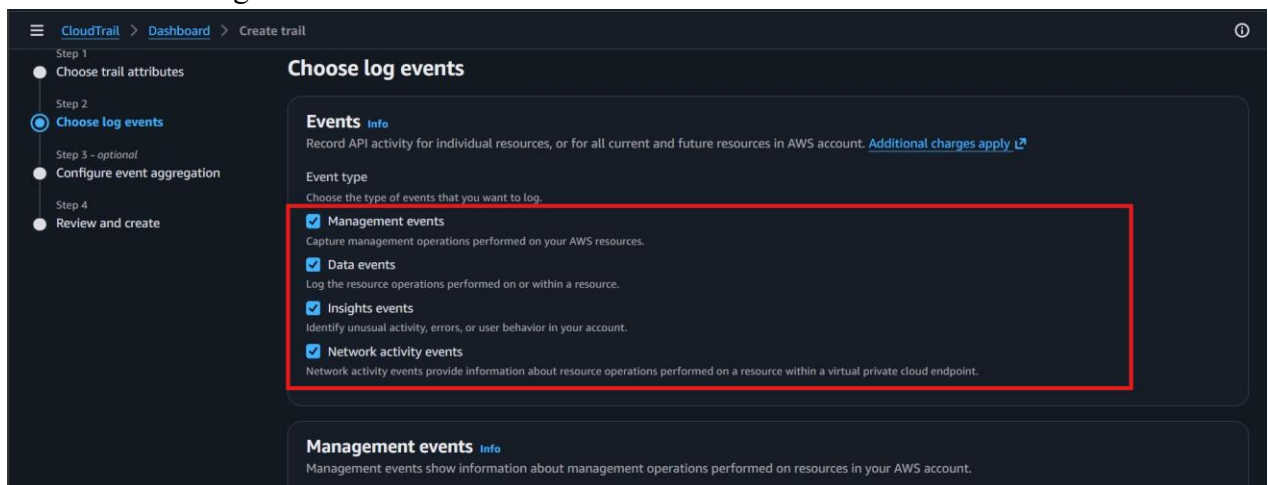
- Enable Log File Validation.
- Keep the remaining default settings.



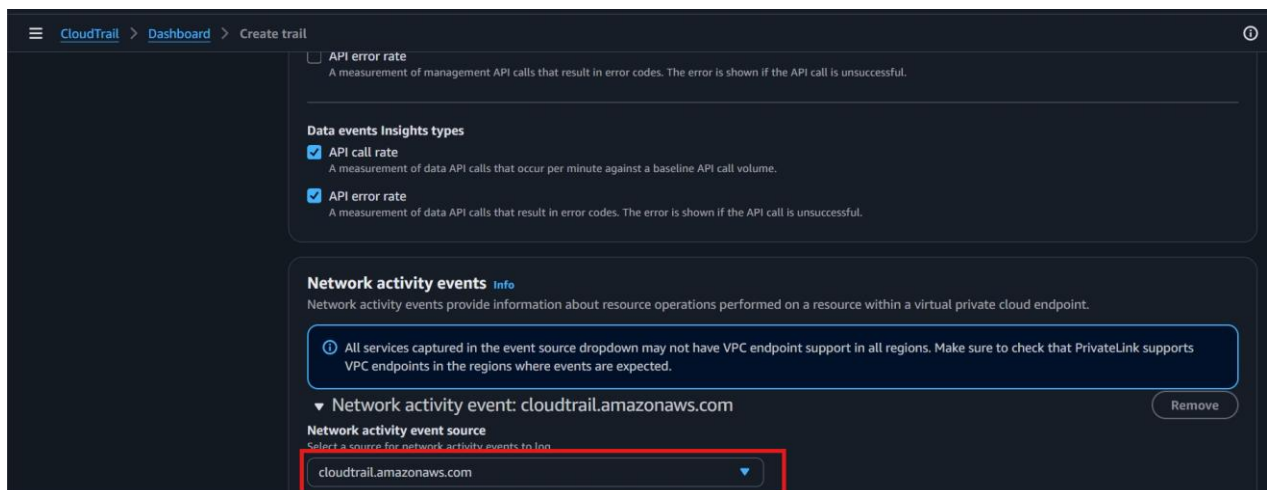
- Click Next.

Step 10 – Configure Event Selection

- Select Management Events.

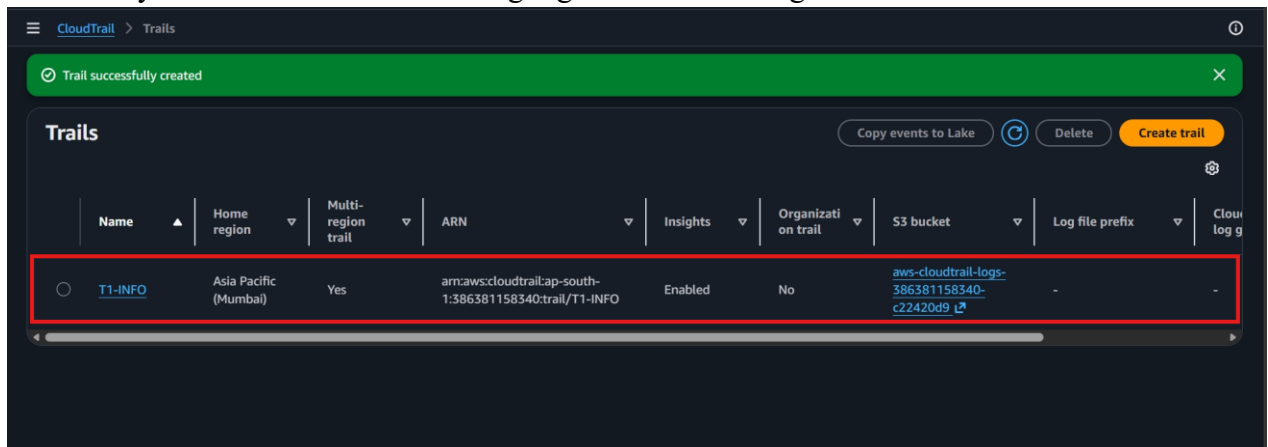


- Keep both Read and Write API events enabled.
- Review the configuration.
- Click Create Trail.

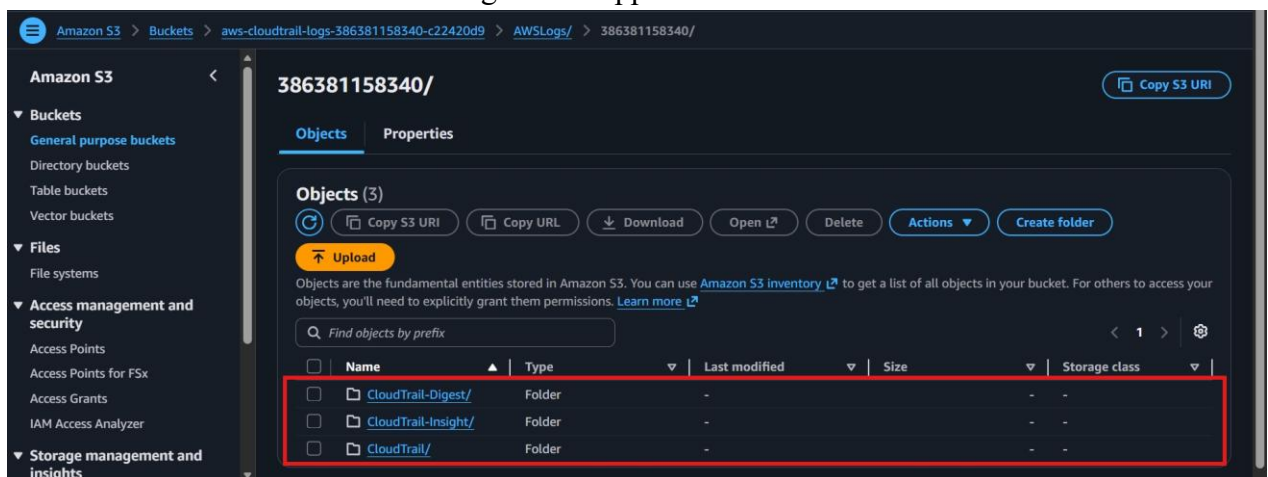


Step 11 – Verify the Trail

- Open the newly created trail.
- Verify that CloudTrail is delivering log files to the configured Amazon S3 bucket.



- Wait several minutes for new log files to appear.



Verification

- An IAM user with Amazon EC2 Full Access was created successfully.
- An EC2 management action was performed using the IAM user.
- The activity was successfully recorded in CloudTrail Event History.
- Event details such as the IAM user, event name, source IP address, event time, and resource ID were successfully verified.
- A CloudTrail Trail was created successfully.
- CloudTrail was configured to store event logs in an Amazon S3 bucket.
- Log File Validation was enabled.
- CloudTrail logs were encrypted using SSE-S3 by default, with AWS KMS available as an optional encryption method.